Nome: Luiz Felipe Ciantela Machado

Turma: CTII-348

Prontuário: CB1990209

Tarefa Básica 9

Matriz Inversa

(Fotos nas páginas seguintes)

Exercícios 1 e 2:

Tarefa Básica 9

Matriz Inversa.

1-) $A = \begin{bmatrix} x & 1 \\ 5 & 3 \end{bmatrix}$ $B = \begin{bmatrix} 3 & -1 \\ y & 2 \end{bmatrix}$

x é invertido, ficando então $\boxed{x = 2}$

3

5 1 → troca o sinal $\boxed{y = -5}$

Soma $x + y = 2 - 5 \Rightarrow \boxed{x + y = -3} \rightarrow \boxed{x + y = -3}$

↳ Letra C.

$1 + 3k = \boxed{1 + 3k}$

2-) $A = \begin{pmatrix} 1 & 0 & 1 \\ k & 1 & 3 \\ 1 & k & 3 \end{pmatrix} \sim \left| \begin{matrix} 1 & 0 & 1 \\ k & 1 & 3 \\ 1 & k & 3 \end{matrix} \right| \begin{matrix} 1 & 0 \\ k & 1 \\ 1 & k \end{matrix}$ $3 + k^2 - (1 + 3k)$

$k^2 - 3k + 2 = 0 \therefore$

↳ $\det = 0$

$3 + 0 + k^2 = \boxed{3 + k^2}$

$\therefore \overset{a}{k^2} \overset{b}{-3k} \overset{c}{+2} = 0$

$\frac{1}{1} + \frac{2}{2} = 3$

$\frac{1}{1} \cdot \frac{2}{2} = 2$ ②

$\Rightarrow \boxed{k' = 1 \quad k'' = 2}$ $\boxed{k' = 1 \quad k'' = 2}$

↳ Letra C.

Exercícios 3, 4 e 5:

3.) $A = \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix}$ $B = A^{t}$

$\rightarrow \det A = 12 - 10 \Rightarrow \boxed{\det A = 2}$

$A^{-1} = \begin{bmatrix} 4 & -5 \\ -2 & 3 \end{bmatrix} \div 2 \Rightarrow \boxed{A^{-1} = \begin{bmatrix} 2 & -\frac{5}{2} \\ -1 & \frac{3}{2} \end{bmatrix}}$

$\hookrightarrow$ letra C.

4.) $\begin{bmatrix} x & 1 & 2 \\ 3 & 1 & 2 \\ 10 & 1 & x \end{bmatrix} \begin{matrix} x & 1 \\ 3 & 1 \\ 10 & 1 \end{matrix} \neq 0$

$20 + 2x + 3x = \boxed{20 + 5x}$

$x^2 + 20 + 6 = \boxed{x^2 + 26}$

$x^2 + 26 - (20 + 5x) \neq 0$

$x^2 - 5x + 6 \neq 0 \therefore$

$\therefore x^2 - 5x + 6$

$\frac{3}{3} + \frac{2}{2} = 5 \Rightarrow x' = 3$

$\frac{3}{3} \cdot \frac{2}{2} = 6$ $x'' = 2$

$\boxed{V = \{(x \neq 3 \text{ e } x \neq 2)\}}$

$\hookrightarrow$ letra A.

5.) $A = \begin{bmatrix} -1 & -1 & 2 \\ 2 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix} \rightarrow A^{-1} = ?$

Dema $= A + A^{-1}$

$2 + 2 + 2 = \boxed{6}$

$\det A = \begin{vmatrix} -1 & -1 & 2 \\ 2 & 1 & -2 \\ 1 & 1 & -1 \end{vmatrix} \begin{matrix} -1 & -1 \\ 2 & 1 \\ 1 & 1 \end{matrix} \quad 7 - 6 = \boxed{1}$

$1 + 2 + 4 = \boxed{7}$

$A = \begin{bmatrix} -1 & 0 & 1 \\ +1 & -1 & 0 \\ 0 & +2 & 1 \end{bmatrix} \Rightarrow \bar{A} = \underbrace{\begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 2 \\ 1 & 0 & 1 \end{bmatrix}}_{A^{-1}} \div 1$

matriz dema $\begin{bmatrix} -1 & -1 & 2 \\ 2 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 2 \\ 1 & 0 & 1 \end{bmatrix} = \boxed{\begin{bmatrix} 0 & 0 & 2 \\ 2 & 0 & 0 \\ 2 & 1 & 0 \end{bmatrix}} \rightarrow$ letra B.

Exercícios 6 e 7:

6-) A e B → matrizes invertíveis de mesma ordem.

X e $(X \cdot A)^T = B \mid X = ?$ $\quad ((A)^T)^T = A$

$A \cdot A^{-1} = I_n = A^{-1} \cdot A$

$((XA)^T)^T = B^T \Rightarrow XA = B^T$

$\underbrace{XAA^{-1}}_{I_n} = B^T A^{-1} \Rightarrow \boxed{X = B^T A^{-1}}_n$

$\boxed{X = B^T A^{-1}}_n$ → Letra B.

7-) $B = \begin{bmatrix} x \\ y \end{bmatrix}$ e $C = \begin{bmatrix} 4x + 5y \\ 5x + 6y \end{bmatrix}$ inversa de A, tal que $AB = C$

$A \cdot B = C$

$A = \underline{C}$ → $A = \begin{bmatrix} 4x + 5y \\ 5x + 6y \end{bmatrix} \div \begin{bmatrix} x \\ y \end{bmatrix}$

B

$A = \begin{bmatrix} 4 & 5 \\ 5 & 6 \end{bmatrix} \Rightarrow \det A = 24 - 25$

$\boxed{\det A = -1}_m$

Adm 2 $\begin{bmatrix} 6 & -5 \\ -5 & 4 \end{bmatrix} \div (-1) \Rightarrow \boxed{A^{-1} = \begin{bmatrix} -6 & 5 \\ 5 & -4 \end{bmatrix}}_m$ → Letra D

Exercício 8:

D S T Q Q S S

8-) $A = \begin{pmatrix} 2 & k \\ -2 & 1 \end{pmatrix}$ | Somo de k para $\det A = 2\det A^{-1}$

$A = \begin{bmatrix} 2 & k \\ -2 & 1 \end{bmatrix} \Rightarrow 2\det A = 2 - (-2k) \Rightarrow 2+2k$

$2\det A = 2\det A^{-1}$

$2+2k = \frac{1}{2+2k} \Rightarrow (2+2k).(2+2k) = 1$

$\therefore (2+2k).(2+2k) = 1$

$4+4k+4k+4k^2-1=0$

$\underset{a}{4k^2} + \underset{b}{8k} + \underset{c}{3} = 0$

$\Delta = +8^2 - 4.4.3 \Rightarrow \frac{-8 \mp 4}{8} \Rightarrow k' = \frac{-8-4}{8} \Rightarrow \frac{-12}{8} \Rightarrow \boxed{\frac{-3}{2}}$

$\Delta = 64 - 48$

$\Delta = 16 \Rightarrow \sqrt{16} = \boxed{4}$

$\Rightarrow k'' = \frac{-8+4}{8} \Rightarrow \frac{-4}{8} \Rightarrow \boxed{\frac{-1}{2}}$

$k' + k'' = \frac{-3}{2} + \frac{(-1)}{2} \Rightarrow \therefore$

$\therefore \Rightarrow \frac{-3}{2} + \frac{(-1)}{2} \Rightarrow \frac{-4}{2} \Rightarrow \boxed{-2}$ → letra B.

Exercício 9:

9-) A e B → ordem 2; det A ≠ 0 e det B ≠ 0

A-) $(A+B)\cdot(A+B) \Rightarrow \boxed{A^2 - AB + BA - B^2}$

B-) $(A+B)^2 = A^2 + 2 \cdot A \cdot B + B^2$

$(A+B)\cdot(A+B) = A^2 + 2AB + B^2$

$\cancel{A^2} + AB + BA + \cancel{B^2} = \cancel{A^2} + 2AB + \cancel{B^2}$

$AB + BA = 2AB$

$AB + BA = AB + AB$

$AB - AB + BA = AB \Rightarrow \boxed{AB = BA}$

C-) $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow \det A = ad - cb$; $-A = \begin{pmatrix} -a & -b \\ -c & -d \end{pmatrix} = \det -A$ ↳ $ad - cb$

$\frac{\det A}{\det -A} \Rightarrow \frac{ad - cb}{ad - cb} = \boxed{1}$

D-) $B = A^{-1} \Rightarrow$ ou seja, $\det B = \frac{1}{\det A}$

$\det A = \det A$

$\det A^{-1} = \frac{1}{\det A}$

$\boxed{\det B = \frac{1}{\det A}}$